

TRACE

Tracking, Recognition, Analytics & City-wide Traffic Enforcement

Backend Schema

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Compatibility Note: Compatible with PRD v1.1 and TRD v1.2. Neither revision altered any functional requirement, data field, or API contract this schema depends on.

1. Introduction

1.1 Purpose

This document defines the database schema that persists every piece of data required by the functional requirements in the PRD, the modules in the Technical Requirement Document, and the screens and components in the App/Website Flow and UI/UX Brief. No table or field in this document is invented independently — each is traced to a specific upstream requirement or UI element, and each is tagged with the FR-xxx/NFR-xxx ID it supports.

1.2 Conventions

- All table and column names use snake_case.
- Every table has a surrogate primary key named <table>_id (UUID) unless noted otherwise.
- All timestamp columns are stored in UTC as timestamptz and are named with an _at suffix.
- Camera and vehicle positions use PostGIS geography(Point, 4326); road segments use geography(LineString, 4326) where a path geometry is needed for map rendering.
- Foreign keys are named <referenced_table_singular>_id.
- Enum-like fields (status, type, role) are implemented as Postgres CHECK-constrained text columns for prototype simplicity, upgradeable to native enum types without a data migration.

1.3 Schema Grouping

Tables are grouped into six areas that mirror the five-layer architecture plus a cross-cutting access-control area: Camera & Road Network, Perception, Identity & Trajectory, Analytics, Alerts & Blacklist, and Access Control.

2. Entity Relationship Overview

- cameras 1—N vehicle_observations (a camera records many observations).
- cameras 1—N road_edges (a camera is an endpoint of many road-graph edges).
- cameras 1—1 camera_reliability_profile (each camera has one active reliability profile).
- vehicle_observations 1—N ocr_reads (a fused observation is built from many raw per-frame OCR reads).
- vehicle_observations N—N vehicle_observations via identity_matches (pairwise candidate/confirmed matches between observations).
- canonical_vehicles 1—N trajectory_points, and trajectory_points N—1 vehicle_observations (a canonical vehicle's trajectory is an ordered chain of observations).
- canonical_vehicles 1—N anomalies (a vehicle's trajectory may raise zero or more anomaly events).
- anomalies and blacklist_matches both feed 1—N into alerts (alerts are a unified log over both anomaly and blacklist triggers).
- road_edges 1—N segment_stats and 1—N congestion_forecasts (analytics and forecasts are computed per edge/segment).
- users 1—N alerts (via reviewed_by) and 1—N blacklist_entries (via added_by).

3. Camera & Road Network Tables

3.1 cameras

Supports: FR-STR-01, FR-PRD-03, NFR-05.

Column	Type	Constraints	Description
camera_id	uuid	PK	Unique camera node identifier.
name	text	NOT NULL	Human-readable camera/junction name.
location	geography(Point,4326)	NOT NULL	Camera coordinates for GIS map plotting.
zone	text	NOT NULL	Zone/sector label, used for OD matrix grouping (FR-ANL-02).
status	text	CHECK IN ('online','degraded','down')	Live feed status shown on Camera Network Status screen.
last_seen_at	timestampz	NULL allowed	Last successful frame ingestion time.

3.2 camera_reliability_profile

Supports: FR-PRD-03 (camera-aware identity weighting).

Column	Type	Constraints	Description
camera_id	uuid	PK, FK -> cameras	One profile row per camera.

day_ocr_reliability	numeric(4,3)	0–1	Estimated OCR reliability in daylight.
night_ocr_reliability	numeric(4,3)	0–1	Estimated OCR reliability at night.
rain_ocr_reliability	numeric(4,3)	0–1	Estimated OCR reliability in rain.
angle_ocr_reliability	numeric(4,3)	0–1	Estimated OCR reliability at oblique angles.
updated_at	timestampz	NOT NULL	Last time the static profile was updated.

3.3 road_edges

Supports: FR-STR-01, FR-STR-02, FR-STR-04 (road-graph reasoning).

Column	Type	Constraints	Description
edge_id	uuid	PK	Unique road-segment identifier.
from_camera_id	uuid	FK -> cameras, NOT NULL	Origin camera node.
to_camera_id	uuid	FK -> cameras, NOT NULL	Destination camera node.
distance_km	numeric(6,2)	NOT NULL	Road distance between the two nodes.
min_travel_time_s	integer	NOT NULL	Fastest plausible travel time, used for impossible-journey checks.
max_travel_time_s	integer	NOT NULL	Slowest typical travel time, used in forecasting baselines.
speed_limit_kmph	integer	NOT NULL	Posted speed limit for the segment.
path_geometry	geography(LineString,4326)	NULL allowed	Optional path geometry for accurate map rendering.

4. Perception Tables

4.1 ocr_reads

Supports: FR-PER-03, FR-PER-04 (raw per-frame evidence prior to temporal fusion).

Column	Type	Constraints	Description
ocr_read_id	uuid	PK	Unique per-frame OCR read.
observation_id	uuid	FK -> vehicle_observations, NOT NULL	Fused observation this read contributes to.
frame_timestamp	timestampz	NOT NULL	Timestamp of the source frame.

raw_plate_text	text	NOT NULL	Unfused plate text read from this single frame.
confidence	numeric(4,3)	0–1	Per-frame OCR confidence score.

4.2 vehicle_observations

Supports: FR-PER-01, FR-PER-02, FR-PER-04, FR-ID-01 (fused, camera-level vehicle sighting record).

Column	Type	Constraints	Description
observation_id	uuid	PK	Unique fused observation.
camera_id	uuid	FK -> cameras, NOT NULL	Camera that recorded the observation.
track_id	text	NOT NULL	Within-camera tracking ID from ByteTrack, prevents duplicate counting.
captured_at	timestampz	NOT NULL	Timestamp of the observation (track midpoint or entry).
fused_plate_text	text	NOT NULL	Confidence-weighted fused plate result across all track frames.
fused_confidence	numeric(4,3)	0–1	Final fused OCR confidence (FR-PER-04).
vehicle_type	text	NOT NULL	Detected vehicle type (e.g., car, SUV, two-wheeler).
vehicle_colour	text	NOT NULL	Detected vehicle colour.

5. Identity & Trajectory Tables

5.1 identity_matches

Supports: FR-ID-02, FR-ID-03, FR-ID-04, FR-STR-02, FR-STR-04 (the pairwise match record that powers the Evidence Panel in the UI/UX Brief).

Column	Type	Constraints	Description
match_id	uuid	PK	Unique candidate/confirmed match between two observations.
observation_id_a	uuid	FK -> vehicle_observations, NOT NULL	Earlier observation in the pair.
observation_id_b	uuid	FK -> vehicle_observations, NOT NULL	Later observation in the pair.
plate_similarity	numeric(4,3)	0–1	Plate-text similarity component.

ocr_confidence_component	numeric(4,3)	0–1	Weighted OCR confidence component.
type_match	boolean	NOT NULL	Whether vehicle type matched.
colour_match	boolean	NOT NULL	Whether vehicle colour matched.
identity_score	numeric(4,3)	0–1, NOT NULL	Composite Identity Score shown as the Confidence Badge (TRD Section 4.2).
implied_speed_kmph	numeric(6,2)	NULL allowed	Computed implied travel speed between the two observations.
is_impossible_journey	boolean	DEFAULT false	Set true when implied_speed_kmph exceeds the edge's plausible ceiling (FR-STR-04).

5.2 canonical_vehicles

Supports: FR-STR-03 (groups confirmed matches into one trackable identity).

Column	Type	Constraints	Description
canonical_vehicle_id	uuid	PK	One row per resolved vehicle identity.
best_plate_text	text	NOT NULL	Highest-confidence plate text representing this vehicle.
first_seen_at	timestampz	NOT NULL	Earliest observation timestamp in the trajectory.
last_seen_at	timestampz	NOT NULL	Latest observation timestamp in the trajectory.

5.3 trajectory_points

Supports: FR-STR-03, FR-STR-05 (the ordered sequence rendered on the Vehicle Trace map).

Column	Type	Constraints	Description
trajectory_point_id	uuid	PK	Unique point in a reconstructed trajectory.
canonical_vehicle_id	uuid	FK -> canonical_vehicles, NOT NULL	Vehicle this point belongs to.
observation_id	uuid	FK -> vehicle_observations, NOT NULL	Underlying observation for this point.
sequence_no	integer	NOT NULL	Chronological order index within the trajectory.

camera_id	uuid	FK -> cameras, NOT NULL	Denormalized for fast map rendering without a join.
captured_at	timestampz	NOT NULL	Denormalized timestamp for fast chronological display.

6. Analytics Tables

6.1 segment_stats

Supports: FR-ANL-01, FR-ANL-03, FR-ANL-05 (Segment Detail panel and heatmap source data).

Column	Type	Constraints	Description
segment_stat_id	uuid	PK	Unique stat row per edge per time window.
edge_id	uuid	FK -> road_edges, NOT NULL	Road segment this statistic covers.
window_start	timestampz	NOT NULL	Start of the aggregation window.
window_end	timestampz	NOT NULL	End of the aggregation window.
density	integer	NOT NULL	Vehicle count on the segment within the window (FR-ANL-01).
avg_speed_kmph	numeric(6,2)	NOT NULL	Average implied speed within the window.
congestion_status	text	CHECK IN ('free','moderate','congested')	Derived bottleneck flag (FR-ANL-03).

6.2 od_matrix_cache

Supports: FR-ANL-02 (precomputed for fast OD Matrix screen rendering).

Column	Type	Constraints	Description
od_entry_id	uuid	PK	Unique origin-destination cache row.
origin_zone	text	NOT NULL	Zone label matching cameras.zone.
destination_zone	text	NOT NULL	Zone label matching cameras.zone.
window_start	timestampz	NOT NULL	Aggregation window start.
window_end	timestampz	NOT NULL	Aggregation window end.
trip_count	integer	NOT NULL	Number of trajectories observed from origin to destination in the window.

6.3 congestion_forecasts

Supports: FR-PRD-01 (Forecast sub-panel on Segment Detail).

Column	Type	Constraints	Description
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forecast_id	uuid	PK	Unique forecast record.
edge_id	uuid	FK -> road_edges, NOT NULL	Segment being forecast.
forecast_for_window	timestampz	NOT NULL	Target time window the forecast applies to.
predicted_density	integer	NOT NULL	Projected vehicle count.
congestion_probability	numeric(4,3)	0–1	Projected likelihood of congestion, shown in the UI as a labelled heuristic estimate.
generated_at	timestampz	NOT NULL	When this forecast was computed.

7. Alerts & Blacklist Tables

7.1 blacklist_entries

Supports: FR-ALT-01 (Blacklist Management screen).

Column	Type	Constraints	Description
blacklist_id	uuid	PK	Unique watched-plate entry.
plate_text	text	NOT NULL	Plate number being watched.
reason	text	NOT NULL	Reason the plate was flagged.
added_by	uuid	FK -> users, NOT NULL	Admin who added the entry.
added_at	timestampz	NOT NULL	When the entry was added.
active	boolean	DEFAULT true	Whether the entry is currently enforced.

7.2 anomalies

Supports: FR-PRD-02 (impossible journeys, duplicate plates, camera-timestamp inconsistencies).

Column	Type	Constraints	Description
anomaly_id	uuid	PK	Unique anomaly event.
type	text	CHECK IN ('impossible_journey', 'duplicate_plate', 'camera_inconsistency')	Anomaly classification shown by icon on the Alerts screen.
canonical_vehicle_id	uuid	FK -> canonical_vehicles, NULL allowed	Related vehicle identity, if applicable.

match_id	uuid	FK -> identity_matches, NULL allowed	Related match record, if the anomaly is match-derived.
details	jsonb	NOT NULL	Structured explanation (e.g., implied speed vs. threshold) for the Evidence Panel.
detected_at	timestamp z	NOT NULL	When the anomaly was raised.

7.3 alerts

Supports: FR-ALT-02, FR-ALT-03, FR-ALT-04 (unified alert log behind the Alerts screen).

Column	Type	Constraints	Description
alert_id	uuid	PK	Unique alert.
type	text	CHECK IN ('blacklist_hit','impossible_journey','duplicate_plate','camera_inconsistency')	Alert type, mirrors anomaly type plus blacklist_hit.
plate_text	text	NOT NULL	Plate the alert concerns.
camera_id	uuid	FK -> cameras, NOT NULL	Camera where the triggering event occurred.
anomaly_id	uuid	FK -> anomalies, NULL allowed	Linked anomaly, when type is anomaly-derived.
blacklist_id	uuid	FK -> blacklist_entries, NULL allowed	Linked blacklist entry, when type is

			blacklist_h t.
triggered_a t	timestamp tz	NOT NULL	When the alert fired.
reviewed	boolean	DEFAULT false	Review status toggle shown on the Alert Card.
reviewed_ by	uuid	FK -> users, NULL allowed	User who reviewed the alert.
reviewed_ at	timestamp tz	NULL allowed	When the alert was reviewed.

8. Access Control Tables

8.1 users

Supports: NFR-06 (role-restricted access).

Column	Type	Constraints	Description
user_id	uuid	PK	Unique platform user.
name	text	NOT NULL	Display name.
email	text	UNIQUE, NOT NULL	Login identifier.
password_hash	text	NOT NULL	Hashed credential, never stored in plaintext.
role	text	CHECK IN ('operator', 'analyst', 'admin')	Determines visible navigation items per the UI/UX Brief.
created_at	timestamptz	NOT NULL	Account creation time.

9. Indexing and Performance Notes

Supports: NFR-02 (interactive-speed trajectory queries).

- vehicle_observations: index on (camera_id, captured_at) to support fast per-camera time-window scans used by segment_stats aggregation.
- identity_matches: index on (observation_id_a), (observation_id_b), and identity_score, to accelerate both trajectory building and confidence filtering.
- trajectory_points: index on (canonical_vehicle_id, sequence_no) to serve the Vehicle Trace screen's ordered list directly without re-sorting.
- road_edges: index on (from_camera_id, to_camera_id) to accelerate reachability checks (FR-STR-02).

- alerts: index on (reviewed, triggered_at DESC) to serve the default "newest unreviewed first" Alerts screen view.
- PostGIS GIST indexes on all geography columns (cameras.location, road_edges.path_geometry) to accelerate map bounding-box queries.

10. Data Retention Notes

Supports: NFR-06. The prototype retains all observation and trajectory data for the duration of the demo dataset without automated expiry. The schema is designed so that a retention job can later be added — deleting rows from ocr_reads and vehicle_observations older than a configured window, while preserving aggregated segment_stats and od_matrix_cache rows for historical analytics — without changing the schema itself. This mirrors the PRD's Section 3.3 note that a full retention/encryption pipeline is documented as production roadmap rather than built for the prototype.

11. API-to-Table Mapping

Cross-reference against the TRD's API Design Overview (Section 5), confirming every endpoint has a backing table.

Endpoint	Primary Table(s)
GET /vehicles/{plate}/trajectory	canonical_vehicles, trajectory_points, vehicle_observations, identity_matches
GET /analytics/heatmap	segment_stats
GET /analytics/od-matrix	od_matrix_cache
GET /analytics/segments/{id}	segment_stats, road_edges
GET /analytics/forecast/{segment_id}	congestion_forecasts
GET /alerts	alerts, anomalies, blacklist_entries
POST / GET /blacklist	blacklist_entries
/ws/live-updates	vehicle_observations, alerts (change-triggered push)

12. Traceability

Every table in this document maps to one or more FR-xxx/NFR-xxx requirements from the PRD, the module design in the TRD, and the screens/components defined in the App/Website Flow and UI/UX Brief. This schema is the final data-contract layer; the Implementation Plan will sequence which tables and endpoints are built in which milestone, and by which team role.

— End of Backend Schema —